

Model Performance Comparison and Final Model Selection

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Contents

Executive Summary	1
1. Evaluation Framework	1
Performance Metric	1
Validation Strategy	1
2. RMSE Comparison Across Models	2
3. Interpretation of Results	3
4. Model Trade-Off Analysis	3
5. Final Model Selection	3
Recommended Model: Random Forest	3
6. When to Use Other Models	3
7. Conclusion	3

Executive Summary

This document presents a comparative evaluation of four predictive models developed to forecast car seat sales across retail stores:

- Multiple Linear Regression (MLR)
- CART (Classification and Regression Tree)
- Random Forest
- Neural Network

All models were trained using cross-validation and evaluated using Root Mean Squared Error (RMSE).

Based on predictive performance, stability, and business applicability, the **Random Forest** model is selected as the final recommended model.

1. Evaluation Framework

Performance Metric

- RMSE (Root Mean Squared Error)
- Lower RMSE indicates better predictive accuracy
- Sales are measured in hundreds of units

Validation Strategy

- Cross-validation used across all models
 - Ensures fair and consistent comparison
-

2. RMSE Comparison Across Models

```
library(dplyr)

##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##   filter, lag
##
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(knitr)

rmse_table <- data.frame(
  Model = c(
    "Random Forest",
    "Neural Network",
    "Multiple Linear Regression",
    "CART"
  ),
  CV_Folds = c(5, 5, 5, 5),
  RMSE = c(
    5.13,
    5.19,
    5.23,
    5.57
  ),
  Model_Type = c(
    "Ensemble (Tree-Based)",
    "Neural Network",
    "Linear Regression",
    "Decision Tree"
  ),
  Interpretability = c(
    "Medium",
    "Low",
    "High",
    "High"
  )
)

rmse_table <- rmse_table %>%
  arrange(RMSE) %>%
  mutate(Rank = row_number())

kable(
  rmse_table,
  caption = "Cross-Validated RMSE Comparison Across Models"
)
```

Table 1: Cross-Validated RMSE Comparison Across Models

Model	CV_Folds	RMSE	Model_Type	Interpretability	Rank
Random Forest	5	5.13	Ensemble (Tree-Based)	Medium	1
Neural Network	5	5.19	Neural Network	Low	2
Multiple Linear Regression	5	5.23	Linear Regression	High	3
CART	5	5.57	Decision Tree	High	4

3. Interpretation of Results

- Random Forest achieves the lowest RMSE (5.13)
- Neural Network shows competitive performance with lower interpretability
- Multiple Linear Regression provides a strong, transparent baseline
- CART offers interpretability but higher prediction error

4. Model Trade-Off Analysis

Model	Accuracy	Interpretability	Stability	Business Readiness
MLR	Moderate	High	High	High
CART	Low	High	Medium	Medium
Neural Network	High	Low	Medium	Medium
Random Forest	Highest	Medium	High	High

5. Final Model Selection

Recommended Model: Random Forest

- Lowest cross-validated RMSE
- Strong generalization performance
- Robust to noise and outliers
- Captures non-linear relationships and interactions
- Suitable for production deployment

6. When to Use Other Models

- MLR: When interpretability is the top priority
- CART: When simple decision rules are required
- Neural Network: When modeling highly complex non-linear patterns

7. Conclusion

After evaluating all models under a consistent framework, **Random Forest is selected as the final production-ready model**, offering the best balance of predictive accuracy and business applicability.